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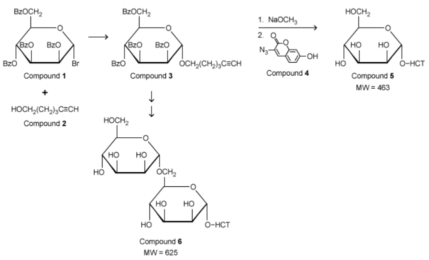

Figure 1 Synthesis of fluorescent coumarin derivatives (Note: HCT denotes the fluorescent butyl hydroxycoumarin triazole group.)

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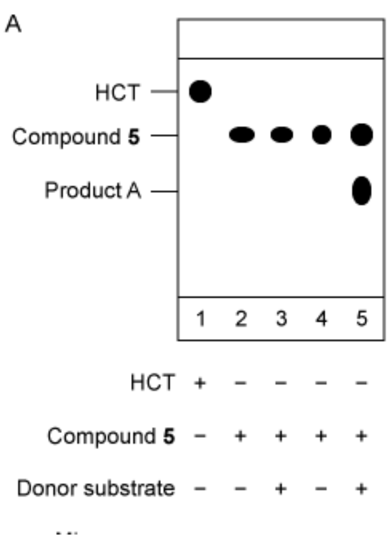


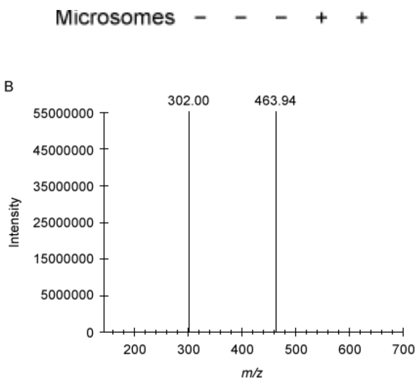


Figure 2 Analysis of Compound **5** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product A fragmented molecular ion peak (*m*/*z* of A prior to fragmentation = 626)

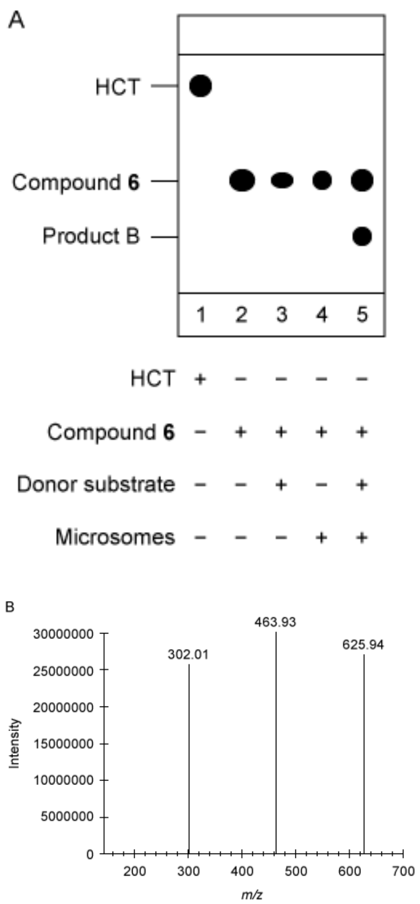


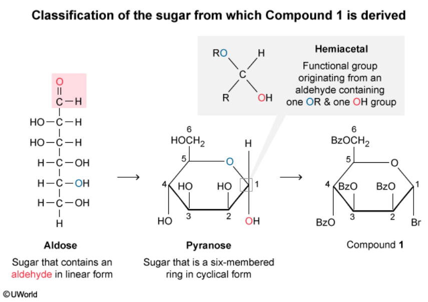
Figure 3 Analysis of Compound **6** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product B fragmented molecular ion peak (*m*/*z* of B prior to fragmentation = 788)

Consider the cyclic form of the sugar from which Compound **1** is derived. Can this sugar be classified as a pyranose and a hemiacetal?

- A. Yes; pyranoses are six-membered rings and hemiacetals are formed from aldoses. [60%]
- B. Yes; pyranoses are five-membered rings and hemiacetals are formed from ketoses. [4%]
- C. No; pyranoses are five-membered rings and hemiacetals are formed from aldoses. [27%]
- D. No; pyranoses are six-membered rings and hemiacetals are formed from ketoses. [6%]

Correct
60% answered correctly

Explanation:



Sugars are linear carbohydrates containing a carbonyl and at least two other carbon atoms bonded to hydroxyl groups. If the carbonyl is an aldehyde (at the end of the chain), the sugar is an **aldose**; if the carbonyl is a ketone (within the chain), the sugar is a **ketose**. A linear sugar can cyclize to adopt either a hemiacetal (aldose) or a hemiketal (ketose) form. A **hemiacetal originates** from an **aldehyde** and consists of a carbon bonded to a hydroxyl group (–OH), an alkyl group, an –OR group (where R is any carbon chain), and a hydrogen atom.

A cyclic sugar can be classified as a **furanose** or a **pyranose** based on its structural similarity to the heterocycles furan or pyran, respectively. Sugars that are five-membered rings (four carbon atoms and one oxygen atom) are furanoses, and sugars that are six-membered rings (five carbon atoms and one oxygen atom) are pyranoses.

Compound **1** is synthesized from a sugar by bromination of C-1 (ie, replacement of the C-1 OH with a Br) and benzoate protection of the OH groups. Because Compound **1** is a six-membered ring containing one oxygen and five carbon atoms, it is **derived from a pyranose**. The sugar from which Compound **1** is derived is also a **hemiacetal** because C-1 on the sugar is bonded to two oxygen atoms (a hydroxyl group and an –OR group), a hydrogen atom, and an alkyl group that **formed from an aldose**.

(Choices B, C, and D) Hemiacetals are formed from aldoses whereas hemiketals are formed from ketoses. Pyranoses are sugars that are six-membered rings whereas furanoses are sugars that are five-membered rings.

Educational objective:
The cyclic structure of a sugar is classified by the size of the ring as well as whether the linear form contains an aldehyde (aldose) or a ketone (ketose). Furanoses are sugars with five-membered rings (four carbons and one oxygen), and pyranoses are sugars with six-membered rings (five carbons and one oxygen). Aldoses cyclize to form hemiacetals, and ketoses cyclize to form hemiketals.

Time spent: 0 sec
Subject: Organic Chemistry

QID: 402471
System: Functional Groups and Their Reactions

The protozoan *Euglena gracilis* expresses glycosyltransferases (enzymes that add carbohydrates to molecules) and glycosidases (enzymes that remove carbohydrates) within its membrane. These enzymes were previously studied using radiolabeled sugars. To avoid the use of radioactive materials, fluorescence-based assays were investigated. Derivatives of the fluorescent molecule coumarin were attached to carbohydrates (Figure 1) for detection in fluorescence assays.

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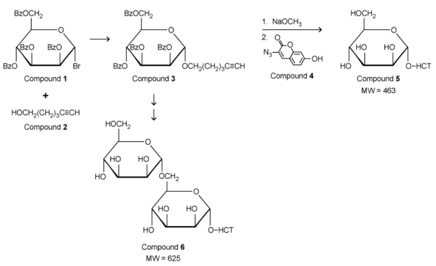

Figure 1 Synthesis of fluorescent coumarin derivatives (Note: HCT denotes the fluorescent butyl hydroxycoumarin triazole group.)

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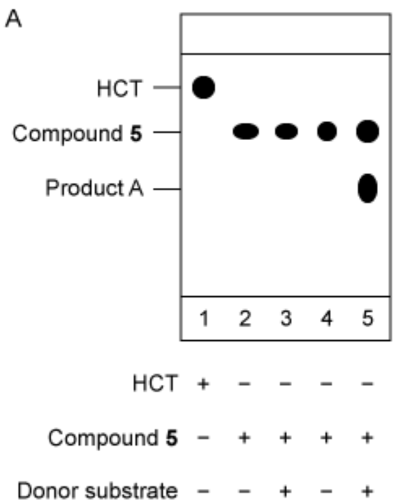


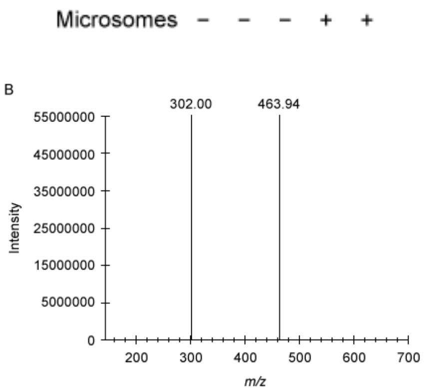


Figure 2 Analysis of Compound **5** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product A fragmented molecular ion peak (m/z of A prior to fragmentation = 626)

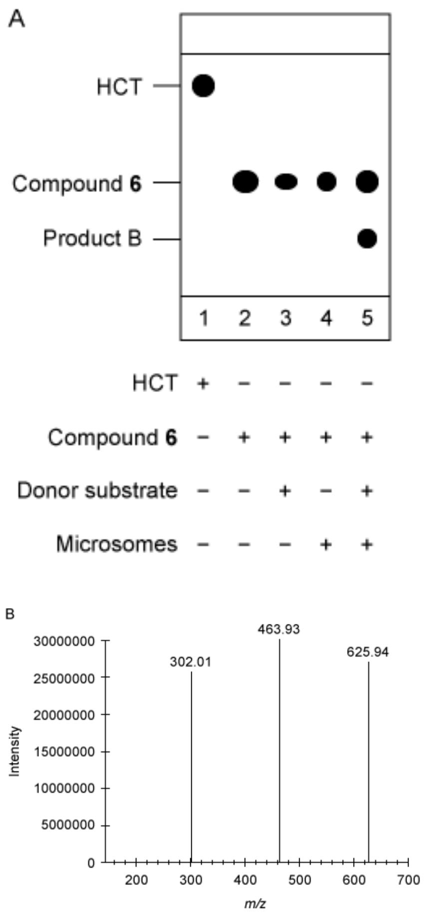
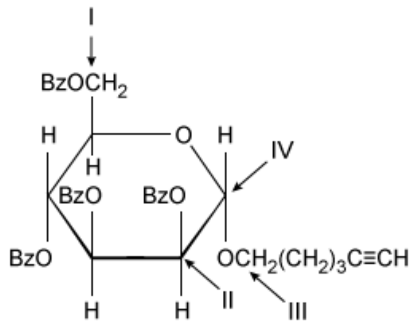


Figure 3 Analysis of Compound **6** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product B fragmented molecular ion peak (m/z of B prior to fragmentation = 788)

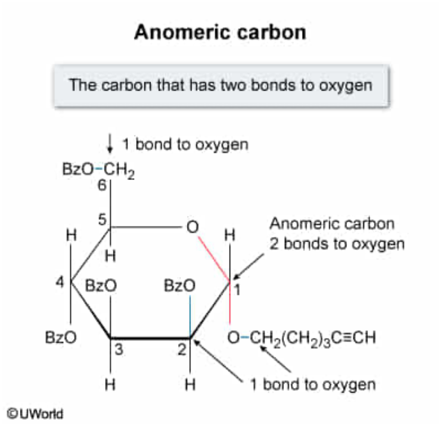


Based on the passage, which carbon gives a distinct C–H coupling in the ^{13}C NMR spectrum that confirms formation of Compound **3**?

- A. I [3%]
- B. II [3%]
- C. III [31%]
- ✔ D. IV [61%]

Correct
61% answered correctly

Explanation:



Sugars can interconvert between linear and cyclic forms. In the linear form, the carbonyl carbon (anomeric carbon) undergoes intramolecular nucleophilic attack by one of the hydroxyl groups on the sugar to cyclize and form a hemiacetal or hemiketal. The **carbonyl oxygen** of the anomeric carbon (C-1 in aldoses and C-2 in ketoses) becomes a hydroxyl oxygen in the cyclic form and can form glycosidic bonds with other biological molecules. In both the linear and cyclic forms, the anomeric carbon is characterized as the carbon in a carbohydrate with **two bonds** to **oxygen**.

The passage states that ^{13}C NMR confirmed that Compound **3** was formed because the anomeric carbon has a distinct C–H coupling in the product. The carbonyl carbon (C-1 in aldehydes and **labeled IV in the question stem**) is bonded to the C-5 hydroxyl oxygen and can be identified as the anomeric carbon because it has two bonds to oxygen.

(Choices A, B, and C) These carbon atoms have only one bond to oxygen and do not correspond to the carbonyl carbon in the linear form. Therefore, they cannot be the anomeric carbon and do not show a distinct C–H coupling in the ^{13}C NMR spectrum.

Educational objective:
The carbonyl carbon (C-1 in aldoses and C-2 in ketoses) of a sugar in linear form and the carbon bonded to two different oxygen atoms in the cyclic form of a sugar is known as the anomeric carbon. This carbon is characterized by two bonds to oxygen and can form glycosidic bonds with biological molecules.

Time spent:0 sec
Subject:
Organic Chemistry

QID:402472
System:
Functional Groups and Their Reactions

The protozoan *Euglena gracilis* expresses glycosyltransferases (enzymes that add carbohydrates to molecules) and glycosidases (enzymes that remove carbohydrates) within its membrane. These enzymes were previously studied using radiolabeled sugars. To avoid the use of radioactive materials, fluorescence-based assays were investigated. Derivatives of the fluorescent molecule coumarin were attached to carbohydrates (Figure 1) for detection in fluorescence assays.

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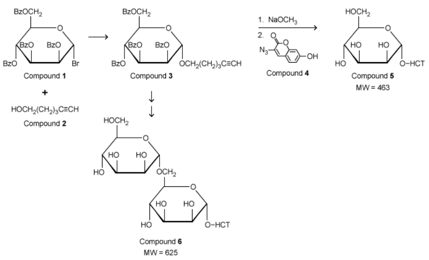

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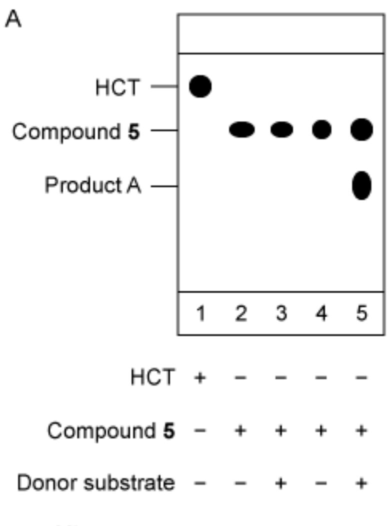

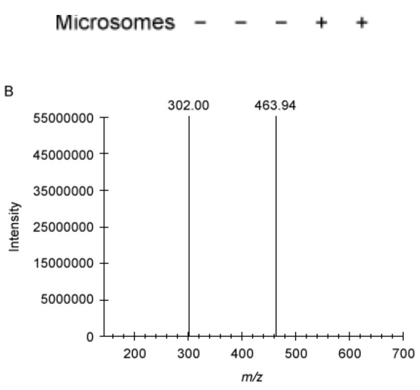

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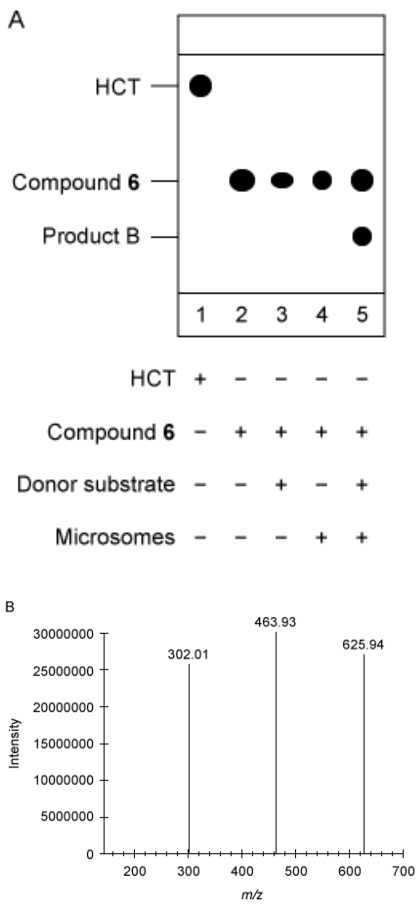


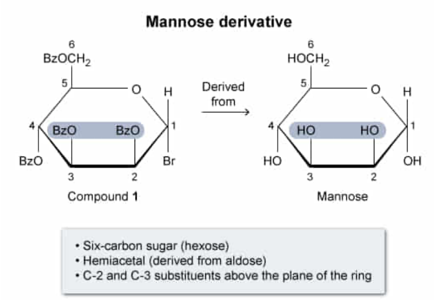
Figure 3 Analysis of Compound **6** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product B fragmented molecular ion peak (*m*/*z* of B prior to fragmentation = 788)

If researchers needed to synthesize Compound **1** before beginning the synthesis of Compound **5**, which of the following sugars would require the fewest steps if used as a starting material?

- A. Fructose [15%]
- B. Ribose [10%]
- ✔ C. Mannose [35%]
- D. Galactose [39%]

Correct
34% answered correctly

Explanation:



Sugars can be **classified** in several ways, including by the stereochemistry of the sugar, the number of carbons in the sugar, the number of carbohydrate units in the molecule, and the functional groups present in the sugar. Some of the most common monosaccharides include ribose, glucose, fructose, mannose, and galactose.

Compound **1** contains six carbons and one of the C-1 substituents is a hydrogen atom, indicating that Compound **1** is derived from a hemiacetal and, more specifically, from a six-carbon aldose. In the orientation shown in Figure 1, the substituents on C-2 and C-3 are both above the plane of the ring and the C-4 substituent is below the plane. These characteristics of Compound **1** match those of **mannose**, which has its hydroxyl substituents in the same orientation. Therefore, mannose can be used to synthesize Compound **1**.

(Choice A) Fructose is a six-carbon ketose with the anomeric carbon at C-2. Compound **1** is derived from an aldose with the anomeric carbon at C-1.

(Choice B) Ribose contains only five carbons, and Compound **1** is derived from a six-carbon sugar.

(Choice D) Galactose differs from Compound **1** in the orientation of the substituents at C-2 and C-4.

Educational objective:
Sugars are classified in a number of ways including by their stereochemistry, the number of carbons in the sugar, the number of sugar units in the molecule, and the type of functional group present in the sugar. Sugars that are prevalent in nature are known by their common names (ie, ribose, glucose, fructose, mannose, galactose).

Time spent:0 sec
Subject:
Organic Chemistry

QID:402473
System:
Functional Groups and Their Reactions

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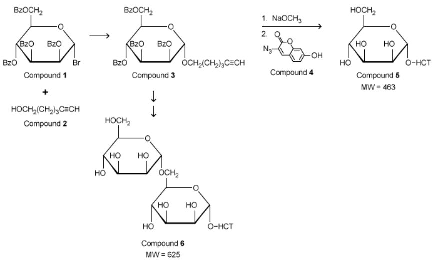

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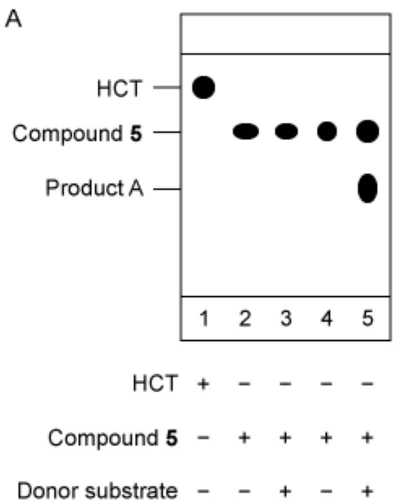

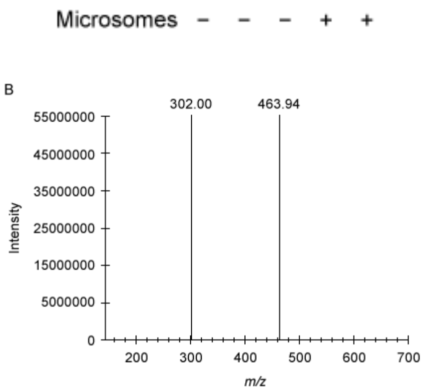

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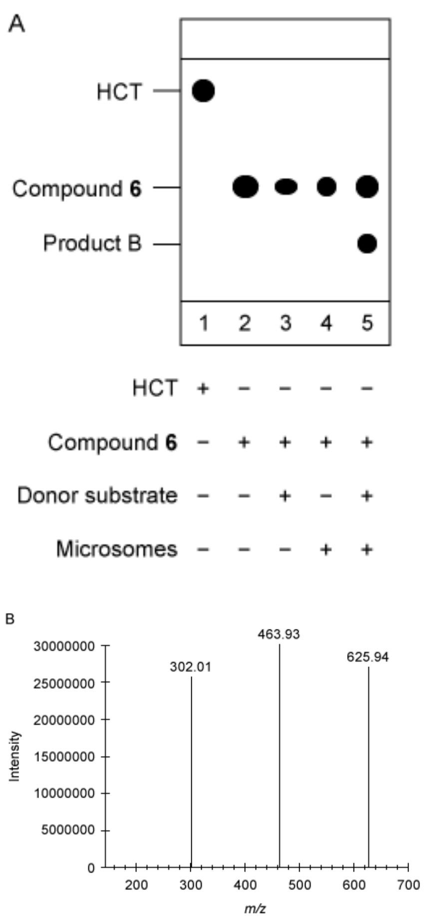


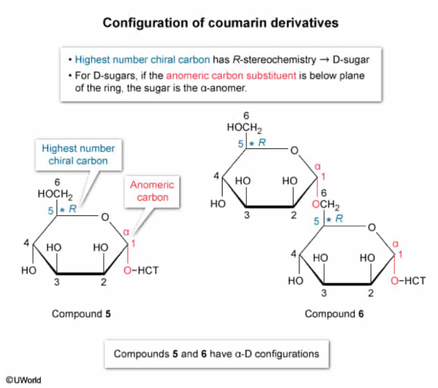
Figure 3 Analysis of Compound **6** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product B fragmented molecular ion peak (m/z of B prior to fragmentation = 788)

What is the configuration of the sugar in the coumarin derivatives shown in Figure 1?

- A. α-L [21%]
- B. α-D [54%]
- C. β-L [11%]
- D. β-D [11%]

Correct
55% answered correctly

Explanation:



Carbohydrates contain chiral centers, and therefore can be classified based on their stereochemistry. Because *R* or *S* designations describe a single chiral center and most sugars have multiple chiral centers, the entire sugar cannot be classified as *R* or *S*. Instead, the carbons in the sugar are numbered, giving the anomeric carbon the lowest number possible (1 in aldoses, 2 in ketoses). The **sugar** is classified as **L or D** based on the **configuration** of the chiral center with the highest number. An *R* configuration at this center designates a D-sugar, and an *S* configuration designates an L-sugar.

Sugars are also classified as α or β based on the configuration of the anomeric carbon. A **cyclic sugar** is in the **α-configuration** if the **anomeric carbon substituent** is on the **opposite side** of the ring plane as the substituent of the highest numbered chiral center; it is in the **β-configuration** if the anomeric substituent is on the **same side** of the plane.

Numbering the carbons in Compounds **5** and **6** starting with the anomeric carbon, C-5 is the chiral center with the highest number. C-5 has an *R* configuration in both compounds, making these compounds D-sugars. Because the hydroxyl substituents of the anomeric carbons are on the opposite side of the ring plane as the C-5 substituents, these compounds are the α-anomers. Therefore, the configuration of Compounds **5** and **6** is α-D.

(Choices A and C) Because C-5 has an *R* configuration, the coumarin derivatives are D-sugars. The L versions would be the enantiomers of the shown configurations.

(Choice D) The substituent of the anomeric carbon would have to be *above* the plane of the ring to be the β-anomer.

Educational objective:
Sugars are classified as L or D based on the configuration of the highest numbered chiral carbon (the anomeric carbon is assigned the lowest number possible). An *R* configuration at the highest numbered stereocenter designates a D-sugar, and an *S* configuration designates an L-sugar. Sugars are also classified as α or β based on the configuration of the anomeric carbon. In α-anomers, the anomeric carbon substituent is on the opposite side of the ring as the highest chiral center and the β-anomer has the substituent on the same side.

Time spent:0 sec
Subject:
Organic Chemistry

QID:402474
System:
Functional Groups and Their Reactions

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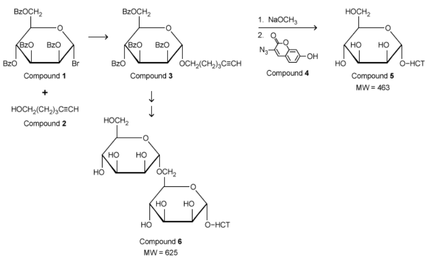

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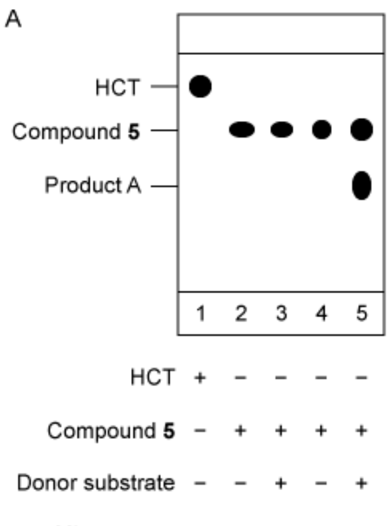

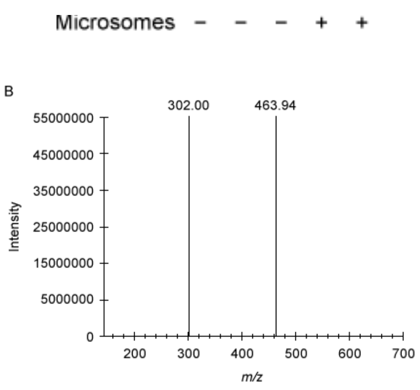

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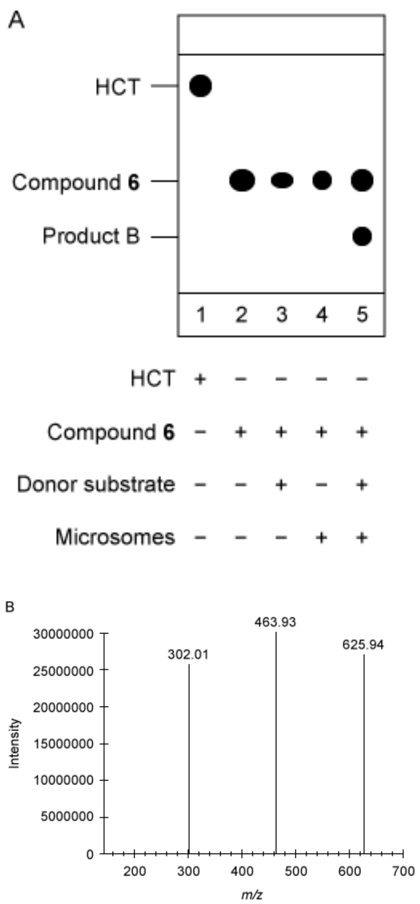


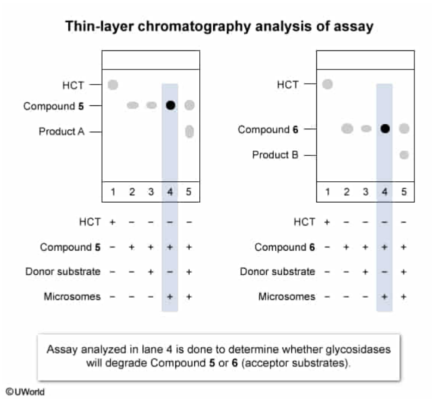
Figure 3 Analysis of Compound **6** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product B fragmented molecular ion peak (*m*/*z* of B prior to fragmentation = 788)

The assays shown in the fourth lane of each TLC analysis in figures 2A and 3A most likely function to:

- A. determine if the enzyme is necessary for glycosyl transfer. [29%]
- B. verify the identity of products A and B. [20%]
- C. demonstrate how HCT affects the donor substrate. [8%]
- D. ensure that glycosidases do not degrade the acceptor substrate. [42%]

Correct
41% answered correctly

Explanation:



The experiment described in the passage explored membrane-bound glycosyltransferase activity in *Euglena gracilis*, and the assays were analyzed by TLC. The glycosyltransferase activity was studied using microsomes, which were isolated from cell membranes. The passage states that membranes also contain glycosidases, which can remove carbohydrates from chains. Therefore, it is necessary to ensure that microsomes do not cause **side reactions** such as degradation of the **reaction substrates**. In the absence of donor substrate, glycosyltransferases in the microsome will not catalyze any reactions. This will allow for conclusions to be drawn about whether the new product in lane 5 is due to glycosidase activity or glycosyltransferase activity.

The assays analyzed in the fourth lanes of the TLC analyses in figures 2A and 3A each contain the acceptor substrate (Compound **5** or **6**, respectively) and the enzyme; the donor substrate was not included. The results from the TLC show that the fluorescent acceptor substrate is still present, but no other fluorescent compounds are present. Therefore, glycosidases did not cleave Compound 5 or Compound 6 to form the new product observed in lane 5.

The assays in the fourth lanes of the TLC analyses function to ensure that **glycosidases do not degrade acceptor substrate**.

(Choice A) To determine whether the enzyme is necessary for the glycosyl transfer, an assay with only the donor substrate and the acceptor substrate (*without* the enzyme) would be necessary. This assay was analyzed in lane 3, not lane 4, on each TLC plate.

(Choice B) Products A and B were not produced in the assay analyzed in lane 4 because the donor substrate was not present, so lane 4 does not help identify these products. Furthermore, the passage says the products were characterized by LC–MS rather than by TLC.

(Choice C) Neither HCT nor the donor substrate is present in the assay analyzed in lane 4.

Educational objective:
To draw conclusions from the results of a study on enzyme activity, control experiments should be done to ensure that no confounding factors, such as side reactions, occur.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 402475
System:
Functional Groups and Their Reactions

The protozoan *Euglena gracilis* expresses glycosyltransferases (enzymes that add carbohydrates to molecules) and glycosidases (enzymes that remove carbohydrates) within its membrane. These enzymes were previously studied using radiolabeled sugars. To avoid the use of radioactive materials, fluorescence-based assays were investigated. Derivatives of the fluorescent molecule coumarin were attached to carbohydrates (Figure 1) for detection in fluorescence assays.

Compound **1**, a carbohydrate derivative with hydroxyl groups protected by benzoate (Bz), reacted with Compound **2** to form Compound **3**. ^{13}C NMR confirmed the formation of Compound **3** by observing a distinct C–H coupling at the anomeric carbon. Hydroxyl deprotection and addition of Compound **4** (a coumarin analogue) gave Compound **5**, a fluorescent coumarin derivative. Compound **6**, another coumarin derivative, was made following a similar sequence except that an additional sugar was attached before the addition of the coumarin group.

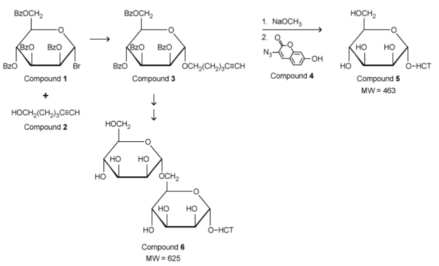
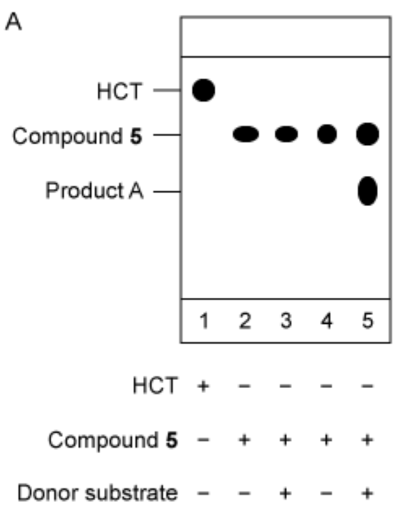


Figure 1 Synthesis of fluorescent coumarin derivatives (Note: HCT denotes the fluorescent butyl hydroxycoumarin triazole group.)

Small vesicles called microsomes were isolated from *Euglena gracilis* membranes, and microsomal glycosyltransferase activities were investigated using a hexose donor substrate and either Compound **5** (Figure 2) or Compound **6** (Figure 3) as the acceptor substrate. Thin-layer chromatography (TLC) analysis indicated the formation of a new product in each assay. These products were purified and characterized by liquid chromatography–mass spectrometry (LC–MS). The molecular ion for each product was then subjected to another round of MS, in which individual hexose units (MW = 162) were cleaved from the coumarin derivatives that were formed during the assay. The MW of hydroxycoumarin triazole (HCT) is 301.



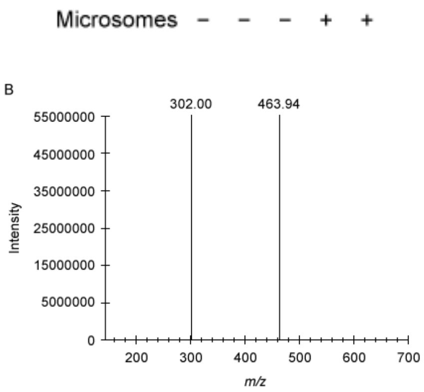


Figure 2 Analysis of Compound **5** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product A fragmented molecular ion peak (*m*/*z* of A prior to fragmentation = 626)

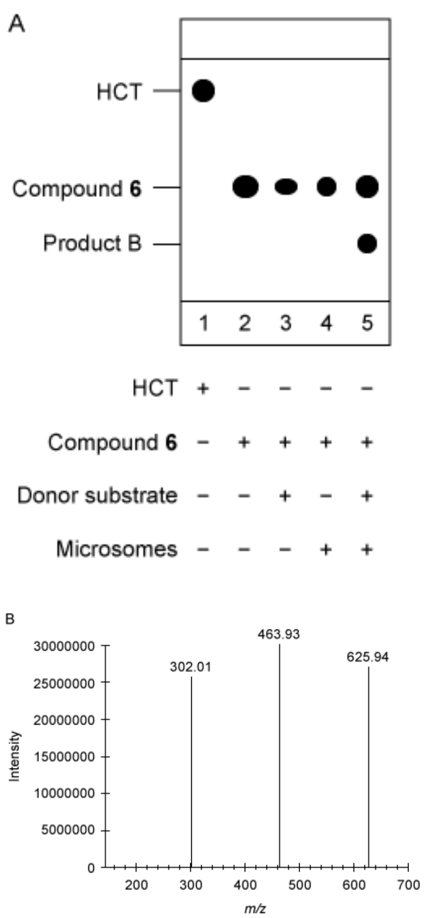


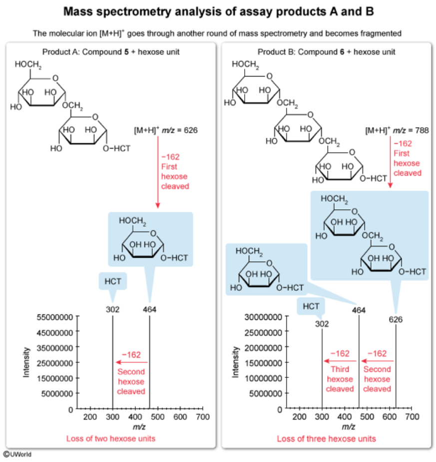
Figure 3 Analysis of Compound **6** as the acceptor substrate: (A) TLC analysis of fluorescent products; (B) Mass spectrum of product B fragmented molecular ion peak (*m*/*z* of B prior to fragmentation = 788)

If the peak with the highest *m/z* ratio on each mass spectrum given in the passage represents the fragment produced after one hexose unit was cleaved from the molecular ions of products A and B, how many total hexose units were cleaved from products A and B, respectively?

- A. two hexose units from product A and two hexose units from product B [11%]
- B. two hexose units from product A and three hexose units from product B [74%]
- C. three hexose units from product A and two hexose unit from product B [10%]
- D. three hexose units from product A and three hexose units from product B [3%]

Correct
73% answered correctly

Explanation:



Mass spectrometry (MS) is a technique that ionizes a molecule and detects the ion's molecular weight. The molecular ion formed is generally the result of the loss of an electron (M^+) or the formation of an adduct with H^+ or Na^+ ($[M+H]^+$ or $[M+Na]^+$). The molecular ion may also break into fragments, resulting in particles with smaller molecular weights. The ions are then detected and a plot (mass spectrum) of the abundance of each ion vs. its mass-to-charge ratio (*m/z*) is generated. The resulting **mass spectrum** can be used to **identify** the masses of a molecule's **fragments** by determining the difference between the *m/z* ratios of selected peaks.

The passage states that after purification of the products A and B by LC–MS, the $[M+H]^+$ of each was run through another round of MS. The question states that the peaks with the highest *m/z* in each mass spectrum represent the fragment resulting from the cleavage of one hexose unit from each $[M+H]^+$. The *m/z* difference between the major peaks in each mass spectrum is 162, corresponding to a single hexose unit.

The mass spectrum for the $[M+H]^+$ of product A has two major *m/z* peaks (464 and 302), indicating one additional hexose unit cleavage to produce the peak at 302. The mass spectrum for the $[M+H]^+$ of product B has *three* major *m/z* peaks (626, 464, and 302), indicating cleavage of *two* additional hexose. Therefore, in total, **two hexose units** were cleaved from product A and **three hexose units** from product B.

(Choice A) The mass spectrum for the $[M+H]^+$ fragmentation of product B indicates the loss of two hexose units, but the question states that the peak with the highest *m/z* is already the result of the loss of one hexose unit. Therefore, a *total of three* hexose units were cleaved from product B.

(Choices C and D) In addition to the hexose unit cleavage described in the question, the mass spectrum for the $[M+H]^+$ fragmentation of product A has *one* pair of peaks with a *m/z* difference of 162, indicating a total of *two* hexose units were cleaved.

Educational objective:
Mass spectrometry is a technique that ionizes and fragments molecules in a sample. The ions are detected, and a plot of ion abundance vs. *m/z* ratio is generated. Fragments of the sample can be identified by the *m/z* difference between two peaks in the mass spectrum.